Math 242, S15
Exam 1

Name: Answer
SSN: xxx-xx-

Instructions:

- There are totally 7 problems.
- You must show all of your work to receive a credit for a correct answer.

Problem	Points	Score
1	13	
2	13	
3	13	
4	13	
5	15	
6	15	
7	18	
Total	100	

Good Luck !

Problem 1. (13 points) Find a general solution of the differential equation:

$$\frac{dy}{dx} = 1 + x^2 - y - x^2 y.$$

(*Hint: Factor the right-hand side to get a seperable equation.*)

Solution:

$$\frac{dy}{dx} = 1+x^2-y-x^2y$$

$$\frac{dy}{dx} = (1+x^2)(1-y)$$

$$\frac{1}{1-y}\,dy = (1+x^2)\,dx$$

$$\int \frac{1}{1-y}\,dy = \int 1+x^2\,dx$$

$$-\ln|1-y| = x + \frac{x^3}{3} + c$$

$$\ln|1-y| = -x - \frac{x^3}{3} - c$$

$$|1-y| = e^{-x-\frac{x^3}{3}+c}$$

$$1-y = A e^{-x-\frac{x^3}{3}}$$

$$y(x) = 1 - A e^{-x-\frac{x^3}{3}}$$

Problem 2. (13 points) Find the particular solution of the following initial value problem:

$$(1+x^2)\frac{dy}{dx} - 2x(1+x^2)y = 2xe^{x^2}, \qquad y(0) = 2.$$

Solution:

$$\frac{dy}{dx} - 2xy = \frac{2x}{1+x^2}e^{x^2}$$

$$P(x) = -2x\,,\quad Q(x) = \frac{2x}{1+x^2}e^{x^2}$$

Integrating factor

$$\rho(x) = e^{\int P(x)dx} = e^{\int -2x\,dx} = e^{-x^2}$$

Thus

$$\frac{d}{dx}[\rho(x)y] = Q(x)\rho(x)$$

$$\rho(x)y = \int Q(x)\rho(x)\,dx$$

$$e^{-x^2}y = \int \frac{2x}{1+x^2}e^{x^2}\cdot e^{-x^2}\,dx$$

$$= \int \frac{2x}{1+x^2}\,dx$$

$$= \ln(1+x^2) + C.$$

$$\Rightarrow y = e^{x^2}(\ln(1+x^2)+c)$$

$$y(0) = 2 \Rightarrow$$

$$2 = e^0(\ln(1) + c)$$

$$\Rightarrow c = 2$$

thus

$$y(x) = e^{x^2}(\ln(1+x^2)+2).$$

Problem 3. (13 points) Find a general solution for the following differential equation:

$$(4x-y)dx+(6y-x)dy=0$$

Solution: $M(x,y)=4x-y$, $N(x,y)=6y-x \Rightarrow \frac{\partial M}{\partial y}=-1=\frac{\partial N}{\partial x} \Rightarrow$ Exact!

$$F(x,y)=\int M(x,y)dx+g(y)$$
$$=\int 4x-y\,dx+g(y)=2x^2-xy+g(y)$$
$$\frac{\partial F}{\partial y}=N \Rightarrow \frac{\partial}{\partial y}(2x^2-xy+g(y))=6y-x.$$
$$-x+g'(y)=6y-x \Rightarrow g'(y)=6y$$
$$\Rightarrow g(y)=3y^2+C_1$$

thus $F(x,y)=2x^2-xy+3y^2+C_1$

and the solution is given by

$$\underline{2x^2-xy+3y^2=C}$$

Problem 4. (13 points) Find a general solution for the following differential equation:

$$2xyy'=x^2+2y^2$$

(*Hint: First transform the equation into the form of homogeneous equation* $y'=F(\frac{y}{x})$)

Solution: $y'=\frac{x^2+2y^2}{2xy}=\frac{1}{2}\left(\frac{x}{y}\right)+\frac{y}{x}$

Let $v=\frac{y}{x} \Rightarrow y=vx \Rightarrow \frac{dy}{dx}=x\frac{dv}{dx}+v$

$$x\frac{dv}{dx}+\cancel{v}=\frac{1}{2}\frac{1}{v}+\cancel{v}$$
$$x\frac{dv}{dx}=\frac{1}{2v} \Rightarrow 2v\,dv=\frac{1}{x}dx$$
$$\int 2v\,dv=\int\frac{1}{x}dx$$
$$v^2=\ln|x|+C$$
$$\Rightarrow \left(\frac{y}{x}\right)^2=\ln|x|+C \Rightarrow y^2=x^2(\ln|x|+C)$$
$$\Rightarrow y=x(\ln|x|+C)^{\frac{1}{2}} \text{ or}$$
$$\underline{y=-x(\ln|x|+C)^{\frac{1}{2}}}$$

Problem 5. (15 points) A certain city has a population of 25000 initially and a population of 30000 after 10 years. Assume that its population grows at a constant rate k per year. Find the population of the city $P(t)$, and how long does it take for the population to reach 40000?

Solution: $\frac{dP}{dt} = kP \Rightarrow P(t) = Ae^{kt}$

$P(0) = 25000 \Rightarrow Ae^0 = 25000 \Rightarrow A = 25000 \Rightarrow P(t) = 25000e^{kt}$

$P(10) = 30000 \Rightarrow 25000e^{10k} = 30000 \Rightarrow e^{10k} = \frac{6}{5}$

$\Rightarrow k = \frac{1}{10}\ln\left(\frac{6}{5}\right) \Rightarrow \underline{P(t) = 25000e^{\frac{t}{10}\ln\left(\frac{6}{5}\right)}}$

$\underline{\approx 25000\,e^{0.01823t}}$

Solve $P(t) = 40000$

$25000\,e^{\frac{t}{10}\ln\left(\frac{6}{5}\right)} = 40000$

$e^{\frac{t}{10}\ln\left(\frac{6}{5}\right)} = \frac{8}{5}$

$\Rightarrow \frac{t}{10}\ln\left(\frac{6}{5}\right) = \ln\left(\frac{8}{5}\right)$

$\Rightarrow t = 10\,\frac{\ln(8/5)}{\ln(6/5)} \approx \underline{25.78 \text{ years}}$

Problem 6. (15 points) Given an initial value problem

$$\frac{dy}{dx} = y + x + 1, \qquad y(0) = -1.$$

Apply **Euler's method** to approximate the solution of the initial value problem on the interval $[0,\ 0.3]$ with step size $h = 0.1$.

Solution: $x_0 = 0$, $y_0 = -1$, $f(x,y) = y + x + 1$, $h = 0.1$.

$x_1 = 0.1$, $y_1 = y_0 + hf(x_0, y_0) = -1 + 0.1\times((-1)+0+1)$

$= -1 + 0 = \underline{-1}$

$x_2 = 0.2$, $y_2 = y_1 + hf(x_1, y_1) = -1 + 0.1\times((-1)+0.1+1)$

$= -1 + 0.01 = \underline{-0.99}$

$x_3 = 0.3$, $y_3 = y_2 + hf(x_2, y_2) = -0.99 + 0.1\times((-0.99)+0.2+1)$

$= -0.99 + 0.1\times 0.21$

$= -0.99 + 0.021$

$= \underline{-0.969}$

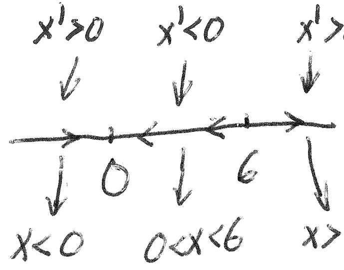
Problem 7. (18 points) Given the following initial value problem

$$\frac{dx}{dt} = 3x^2 - 18x, \qquad x(0) = 5.$$

(a) Find all critical points of the given differential equation and then determine whether it is stable or unstable for each of them.

$f(x) = 3x^2 - 18x = 3x(x-6)$

$3x(x-6) = 0 \Rightarrow x=0,\ x=6$ two critical points



$\underline{x=0 \text{ stable}}$

$\underline{x=6 \text{ unstable}}$

(b) Solve the initial value problem for $x(t)$ and find $\lim_{t\to+\infty} x(t)$.

$\frac{dx}{dt} = 3x(x-6)$

$\int \frac{1}{x(x-6)}\,dx = \int 3\,dt$

$\frac{1}{6}\int \left(\frac{1}{x-6} - \frac{1}{x}\right)dx = \int 3\,dt$

$\ln|x-6| - \ln|x| = 18t + C$

$\ln\left|\frac{x-6}{x}\right| = 18t + C$

$\frac{x-6}{x} = Ae^{18t}$

$x(0) = 5 \Rightarrow \frac{5-6}{5} = Ae^0$

$\Rightarrow A = -\frac{1}{5}$

$\frac{x-6}{x} = -\frac{1}{5}e^{18t}$

$1 - \frac{6}{x} = -\frac{1}{5}e^{18t}$

$\frac{6}{x} = 1 + \frac{1}{5}e^{18t}$

$\underline{x(t) = \frac{6}{1+\frac{1}{5}e^{18t}}}$

$\lim_{t\to+\infty} x(t) = \lim_{t\to+\infty} \frac{6}{1+\frac{1}{5}e^{18t}}$

$= \underline{0}$